The 3n+1 Machine: Interactive Visual Orbit Explorer, Stopping-Time Analyzer, and Parity-Vector Platform for Collatz Dynamics

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ORCID: 0009-0006-7068-3718 | GitHub: BenFrohman/the-3n-plus-1-machine
Dedicated 1-to-1 Zenodo DOI: 10.5281/zenodo.23653011

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Abstract

Standalone TypeScript/React/Node computational platform (‘src/‘, ‘server/‘, ‘shared/‘, ‘migrations/‘) for interactive exploration of 3n+1 Collatz trajectories, 2-adic parity-vector trees, Terras stopping-time distributions, and cycle-equation invariants.

1 Scope, Scientific Honesty, and Non-Claim Boundary

Computational and visual exploration platform for verified finite-range Collatz dynamics and algebraic cycle identities; does not claim an unconditional proof of the global 3n+1 conjecture.

- **Repository:** `https://github.com/BenFrohman/the-3n-plus-1-machine`
- **Dedicated 1-to-1 Zenodo DOI:** `10.5281/zenodo.23653011`
- **Earliest Git Commit SHA:** `ce2663622c04171e4273646bd8520aefe9341d7d` (2026-09-18 04:27:42 PM CDT (UTC-5))
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2 Technical Overview & Repository Specification

A cinematic visual proof of almost everything that is known about the Collatz conjecture.
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Open source under the Apache License, Version 2.0. See LICENSE.
The paid-use restriction of 23 September 2026 was a mistake and is withdrawn.
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Author	Benjamin Stanley Frohman
Version	‘1.1.0‘
Status	Public · Apache-2.0
First fixed	18 September 2026
License	Apache-2.0
An interactive film of the ‘3n+1‘ map:
- the rule

- the hailstone of 27 (peak 9232)
- the reverse coral tree
- Terras density
- the parity-vector engine
- the remaining lemma (Tao 2019, machine check below \{}(2^{68}\{}), no other cycle proven)

3 Cryptographic Provenance & File Inventory

Every source artifact in `BenFrohman/the-3n-plus-1-machine` is audited via `titan-audit` (`audit/TITAN_AUDIT_CERTIFICATE.json`) and sealed under a deterministic SHA-256 Merkle tree via `sealguard` (`PROVENANCE_MERKLE_ROOT.json`) signed with Ed25519 key fingerprint `b428fb72676f1c52`.

Repository Path	Size	SHA-256 Prefix
.github/CODEOWNERS	14 B	db51a51ebf7d5900...
.github/RELEASE.md	648 B	e22ea0a920cee25e...
.gitignore	194 B	6ab49d54e18c2caa...
.grok/app-env.json	43 B	0a7458baafdc2464...
.npmrc	48 B	5af3d474eb1886c0...
.prettierrc	90 B	d5f90f76c5b68ce7...
CITATION.cff	832 B	ad1060bae44a6a8b...
COPYRIGHT	1199 B	af2624f4181f4bd3...
LICENSE	11355 B	9da72c2bee4f20ba...
NOTICE	1025 B	d5f21cae93516f1c...
README.md	1795 B	3dc3a94e5188b472...
SECURITY.md	476 B	7d1b689661ceef62...
eslint.config.mjs	1352 B	a840049ae76404c6...
migrations/0001_auth.sql	2739 B	f953cacc448c0c81...
migrations/0002_orbits.sql	938 B	8f74f1caf71a28a0...
migrations/auth/0001_auth.sql	2739 B	f953cacc448c0c81...
package-lock.json	262769 B	fc81314ce9ca4b6b...
package.json	4332 B	9337efc19955689d...
public/favicon.svg	399 B	6de8eddf1d851737...
public/og.jpg	94780 B	2619129a9d8811c0...
public/x-banner.jpg	61576 B	24d51a5d09a9971e...
scripts/app-env-plugin.mjs	1353 B	c9012cb367bd0d1b...
scripts/brand-check.mjs	9684 B	31c56f74d8461e2e...
scripts/brand-check.test.mjs	15159 B	fa2941f008c105fb...
scripts/browser-guard.mjs	1799 B	ce18f320c69f8950...
scripts/browser-smoke-verdict.mjs	4738 B	cf9679b8a4dd7136...
scripts/browser-smoke-verdict.test.mjs	15718 B	5ff069af0c316480...
scripts/browser-smoke.mjs	6117 B	ea272cd39ef58512...
scripts/check-auth-invariant.mjs	4278 B	3c60e6a4dc64749f...
scripts/check-auth-invariant.test.mjs	4293 B	e5146edf5b2d5fd3...
scripts/grok-pwa-plugin.mjs	6690 B	19ea58acb8752be4...
scripts/grok-pwa-plugin.test.mjs	20395 B	9770f1be1797ecd5...
scripts/grok-pwa-shared.d.mts	3144 B	99c8342319978d71...
scripts/grok-pwa-shared.mjs	17750 B	5a888cba25e4cff0...
scripts/install-page.html	5739 B	f02e277ec9daec42...

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